



M002 Week 8 — English Worksheet

Topic: Capstone — 20×20×20 Cube Maze · **Course:** Maze Madness · **Time:** about 60 minutes

This is the **final project**. You design a 3D cube maze with redstone doors and levers, then write code that solves it on its own.

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
A cube maze: 20 wide, 20 deep, 20 tall.
- multiple floors
- at least 1 redstone door
- at least 1 lever
- a clear start and finish
```

Sketch (in words) one floor of your maze. Where is the start? Where is the goal?


```
keep doing until at goal:
  if wall ahead OR closed door ahead:
    turn right
  otherwise if lever ahead:
    interact ahead
  otherwise:
    move forward
```

This is one strategy. Name one type of maze where this will get stuck.


```
function go_to_next_floor:
  while no stairs ahead:
    move forward
  move up by 1
```

Why is it useful to have a function called `go_to_next_floor` in a 3D maze?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent must use `interact` only when there is a **lever** ahead — not on every step.

```
keep doing until at goal:
    interact ahead
    if wall ahead:
        turn right
    otherwise:
        move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent should **stop** when it reaches the goal. Right now it keeps running forever.

```
keep doing forever:
    if wall ahead:
        turn right
    otherwise:
        move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — On a 3D maze the agent must also handle **going up to the next floor**. Right now the code only handles left/right/forward.

```
if wall ahead:
    turn right
otherwise:
    move forward
```

Write the fixed code (add an "if stairs ahead" or similar):

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now build your **final project**. Your maze must include:

- a **20 × 20 × 20** cube structure
- **more than one floor**
- at least **1 redstone door**
- at least **1 lever** the agent must flip
- a clear **start** and **finish**
- agent code that solves it on its own using `while` + `if/else` + `interact`

When you finish, come back here. Send a photo or video of the agent solving your maze, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

My maze has ... floors and ... levers.

The agent's strategy is ...

The hardest part of building the maze was ...

The hardest part of writing the code was ...

To fix it, I ...

The thing I am most proud of is ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while the agent runs your maze. Talk through it like you are teaching someone who has never seen it. Try to use these words: **cube, floor, lever, door, strategy**.

1. Show the cube maze from outside, then from inside.
2. Point out the start, the goal, the door, and the lever.
3. Run your code and follow the agent all the way to the goal.
4. Read the most important loop in your code out loud and say what it does.
5. Say what you learned across the 8 weeks of this course.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk. This is the **last week** of Maze Madness — congratulations on finishing the full Minecraft track!